

Reconfigurable Roadways: Envisioning the Future of V2X-Driven Autonomous Urban Infrastructure through an Interactive Tabletop Simulator

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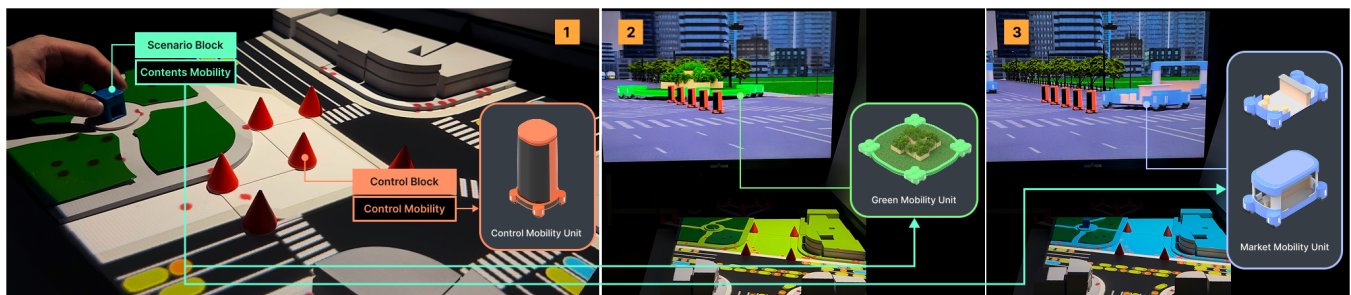


Figure 1: Operational Tabletop and Scenario Video with System Blocks and Units

Abstract

Autonomous driving and cooperative platooning are expected to transform urban mobility, significantly reducing traffic density in populated cities. Therefore large portions of road infrastructure could become underutilized, presenting valuable opportunities for spatial reallocation. These spaces could host Tactical Urbanism strategies supporting socially adaptive, community-centered urban transformation. We introduce Reconfigurable Roadways (RR)—an adaptive system reallocating road space by expanding pedestrian zones and dynamically modulating vehicle flows through real-time demand. Using V2X-enabled traffic coordination, RR offers a scalable framework for highly flexible, multi-use streets. To illustrate RR and clarify current technology and diverse future possibilities, we present an interactive tabletop simulator of an urban mobility scene. The tangible simulator lets users rearrange modular blocks—each representing a roadway role—to explore near-future scenarios and

observe simulated changes in traffic behavior and urban form. The platform fosters intuitive understanding of RR principles and engages users in shaping future urban mobility systems.

CCS Concepts

• **Human-centered computing** → **Systems and tools for interaction design.**

Keywords

Reconfigurable roadway, Autonomous driving, V2X, Tactical urbanism

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1 Introduction

Population decline and the commercialization of advanced Vehicle-to-Everything (V2X) autonomous driving systems are poised to fundamentally reshape urban roadways— from mere corridors of

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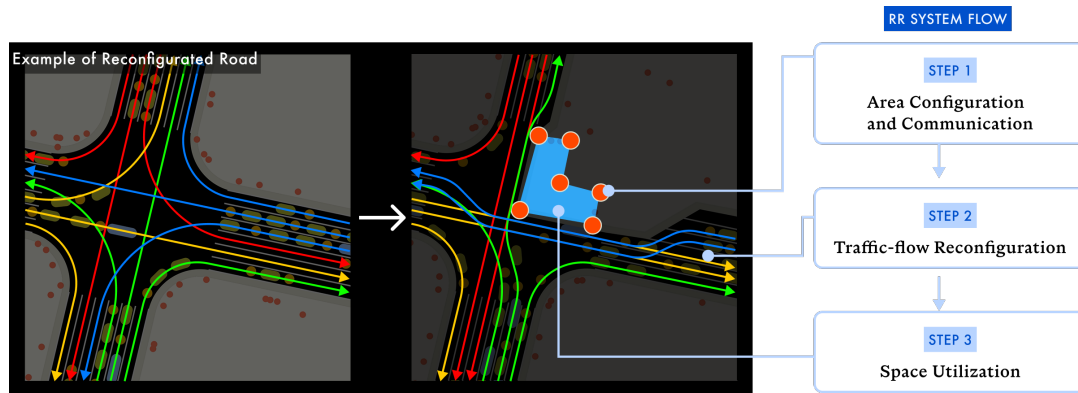


Figure 2: Example of Traffic Flow Changes with RR System Applied and the System Operating Sequence

movement into dynamic infrastructure capable of generating surplus urban space. Recent large-scale network simulations of metropolitan traffic demonstrate that congestion can be cut to one-quarter of its current intensity, while the average flow speed of autonomous and connected vehicles can exceed that of human-driven traffic by up to 300% [3]. These shifts imply that the same physical infrastructure could support more vehicles with significantly shorter dwell times, unlocking vast amounts of underutilized space—such as lanes, parking zones, and buffer areas—for reimagined civic use. To address this emerging opportunity, we introduce the concept of Reconfigurable Roadway (RR)—an infrastructure paradigm capable of adapting in real-time to evolving urban demands and transportation contexts, once autonomous vehicles (AVs) become ubiquitous. Leveraging real-time V2X data alongside tactical urbanism principles, RR repurposes latent roadway capacity into flexible public assets, turning idle pavement into programmable zones for pedestrian activity, cultural events, or commercial uses on demand. To foster experiential exploration and intuitive understanding of this concept, we developed an interactive tabletop simulation platform. Users manipulate physical blocks on a scaled urban model to close or reallocate lanes, or to insert new urban functions, while the system instantly projects V2X-driven traffic simulations directly onto the tabletop model, allowing them to observe the dynamic impact of their interventions. This setup allows them—regardless of technical background—to intuitively explore and reflect on reconfigurable roadway scenarios, serving as a learning and engagement platform fostering broader acceptance of adaptive infrastructure strategies in the transitional era of autonomous urban mobility. The primary contributions of this study are as follows:

- Strategic framework for idle-space utilization: We propose a strategic model for reimagining idle road space in the era of autonomous driving as dynamically reconfigurable infrastructure that adapts to evolving urban needs.
- Interaction-based implementation for conceptual engagement: We developed a tangible tabletop simulator integrating real-time visualization, enabling intuitive understanding of RR concept and its accompanying future mobility technologies.
- Establishing a foundation for social acceptance and discourse: User feedback indicated greater receptiveness to future change

and demonstrated the potential to broaden participatory design and civic dialogue.

2 BACKGROUND

2.1 Platooning System

Platooning is a cooperative autonomous-driving technology that binds vehicles into tight formations by transmitting the lead vehicle's motion data via vehicle-to-vehicle (V2V) links and keeping short headways through Cooperative Adaptive Cruise Control (CACC). By coordinating control through shared motion data rather than individual sensors, it reduces reaction delays and enables high-precision collective driving. Studies [4, 7] show that CACC reduces traffic shockwaves and improves flow stability, effectively increasing throughput. When widely adopted, it can double lane capacity [4] without expanding road infrastructure—potentially creating surplus road space.

2.2 Tactical Urbanism

Tactical Urbanism proposes deliberate, flexible alternatives to traditional one-size-fits-all urban development through short-term, low-cost, and scalable interventions [2]. These approaches allow for collecting immediate user feedback, rapid iterative improvement—often leading to the permanent adoption of successful temporary installations. For example, the New York City Department of Transportation transformed a provisional pedestrian plaza into a permanent public space in Times Square based on positive public response. Similarly [5], Melbourne's "Playful Parklet" temporarily transformed curbside parking into spaces for play and rest, which users described as "a joyful and meaningful" [6]. While Tactical Urbanism has proven effective in reimagining urban space, it still requires substantial human and material resources. The autonomous platooning technology presented in this study offers a new infrastructural foundation for urban reconfiguration. By implementing the RR, Tactical Urbanism-style interventions can be conducted more efficiently and dynamically, reducing manual overhead and supporting a more sustainable, adaptive, and resilient urban development model.

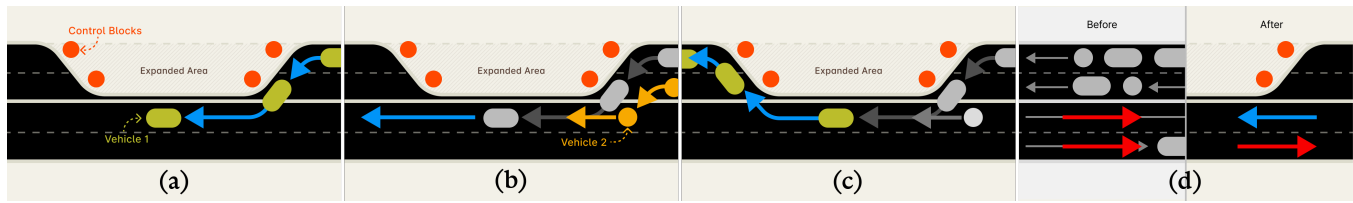


Figure 3: Key Reconfiguration Rules: (a) Rerouting, (b) Merging, (c) Deviation and Return, (d) Lane Reversal

3 DESIGN RATIONALE AND HARDWARE IMPLEMENTATION

3.1 Reconfigurable Roadway (RR) Overview

The core operating mechanism of RR can be summarized as follows:

- (1) Space designation: Deployed Control Mobility units detect each other's positions and define restricted zones between them, broadcasting this information to nearby vehicles via V2X.
- (2) Traffic-flow rerouting: Incoming vehicles detect the placement of Control Mobility units, and adjust their paths accordingly, forming stable traffic stream through platooning.
- (3) Space repurposing: Freed-up lanes converted into pedestrian-friendly or community-oriented spaces (e.g., parks, markets) based on local needs.

By consolidating previously underutilized road sections, RR transforms fixed traffic infrastructure into a flexible, multi-use urban asset.

3.2 Reconfiguration Rules

To integrate platooning-style adaptability into RR, we defined key behavioral rules from both driver and pedestrian perspectives: The vehicle paths within RR are governed by the following dynamic adjustment rules: **Rerouting** (Fig.3(a)): Vehicles are redirected via adjacent lanes when a Control Block closes a segment. **Merging** (Fig.3(b)): Multiple lane streams merge into one, with V2V communication managing speed and spacing. **Deviation and Return** (Fig.3 (c)): Vehicles take a detour around restricted zones and rejoin their original route afterward. **Lane Reversal** (Fig.3 (d)): Lane direction is temporarily reversed when needed (e.g., reallocating one lane to oncoming traffic while redistributing others), allowing for dynamic configuration.

3.3 Citizen-Participatory TUI Platform for Expansive Discourse on Reconfigurable Roadways

To expand public engagement, we introduced a Tangible User Interface (TUI) [1] which lets participants intuitively explore the futures shaped by RR concept, and imagine—then debate—how newly liberated road space might be used. When users place physical blocks on the table, real-time traffic simulations are projected onto the table, instantly visualizing the spatial impact of each configuration. RR's core elements are embodied in physical blocks as follows:

- (1) **Control Blocks** (Fig.1 (1) marked in orange) – Represent Control Mobility units which govern vehicle flow: arranging them restricts vehicle access within the defined area.
- (2) **Scenario Blocks** (Fig.1 (1) marked in cyan) – Represent Contents Mobility units; adding them transforms road space into alternate uses such as parks, markets, or pedestrian zones, allowing users to explore diverse future scenarios.

4 Discussion

We staged public installations for one day for three hours, inviting visitors to interact with the Reconfigurable Roadway (RR) models in an unstructured and natural manner, while observing their engagement. Visitors reported that the tangible experience made the RR concept more credible and inspiring. Emerging ideas included converting selected lanes into pedestrian zones, night-time roads into open lounges, and repurposing idle space for medical hubs in emergencies. Ultimately, the system demonstrated how interactive, physical modeling can foster imagination, broaden discourse, and bridge speculative urban concepts with practical design thinking. To improve the framework and quantitatively evaluate the feasibility of RR, acquiring also insights from professional urban designers in using the platform remains future work.

5 Conclusion

This study reimagines roadway infrastructure in the era of autonomous driving as a reconfigurable civic asset that can adapt to evolving urban demands. To bridge concept and experience, we developed a tabletop interactive system that enables users to physically explore idea. This tangible prototype not only enhances public understanding but also encourages imaginative engagement. User feedback supports the potential of this hybrid approach as a tool for civic discourse, urban planning, and participatory design—laying the groundwork for future integration with real-time data and policy-making processes.

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